Computer Security

Security Foundations and User Authentication

Assignment 1

Prepared by Sajjad Alimohammadi,

Teaching Assistant

Amirkabir University of Technology *(Tehran Polytechnic),*
Department of Computer Engineering

November 2026

1. Core Security Concepts (9 points)

(a) Define computer security and explain how it differs from safety. Define each of the three objectives of computer security (confidentiality, integrity, and availability) and give an example of each.

(b) Define threat, vulnerability, and attack, and use an example to show how the three concepts are related.

(c) Why does the security of a computer system depend on the policy it is required to enforce? In your answer, explain what a secure state and a reachable state are.

2. Grade Change Policy (21 points)

A university grade system allows only the instructor of record to change a grade after completing two-factor authentication. After the deadline, the system also requires recorded approval from the academic office. For this problem, the model contains only the five states below.

State	Description
A	No grade change has been made (initial state).
B	The instructor changes the grade before the deadline after completing two-factor authentication.
C	The instructor changes the grade after the deadline with two-factor authentication and academic office approval.
D	The instructor changes the grade before the deadline using only a password.
E	A teaching assistant changes the grade before the deadline after completing two-factor authentication.

The current system can move from A to B through the regular workflow and from A to D through an emergency workflow. The post-deadline workflow is disabled, and a role check prevents teaching assistants from changing grades. No other transitions are possible in this model.

(a) List the set of secure states S and the set of reachable states R. Classify the current system as secure, precise, or insecure and justify your answer.

(b) The first proposal removes only the emergency workflow. The second also enables a direct transition from A to C under the conditions specified in the policy. For each proposal, give R and determine whether the system is secure and whether it is precise.

(c) For the second proposal, design one test that should allow a grade change and two tests that should deny it. Specify the inputs and expected result for each test. At least one denial test must show that completing the second factor does not by itself authorize a grade change.

3. Biometric Authentication Thresholds (20 points)

To enter a research lab, a user provides an identifier and a fingerprint. The system compares the sample only with the template associated with that identifier. It grants access when the similarity score is at least the decision threshold.

The evaluation used the same 1,000 authorized and 200 unauthorized attempts for each threshold. The policy caps the false acceptance rate at 1% and the false rejection rate at 4%.

Threshold	False acceptances	False rejections
Low	10	10
Medium	2	30
High	0	80

(a) Does the system perform identification or verification? Explain the role of the user-supplied identifier.

(b) For each threshold, calculate the false match and false nonmatch rates and show their denominators. Which threshold satisfies both policy limits?

(c) At the medium threshold, 12 false rejections occurred in a group that made 100 of the 1,000 authorized attempts. Calculate the group's false rejection rate and explain what the overall rate hides.

(d) Propose a fallback for an authorized user repeatedly rejected by the fingerprint check. Specify the evidence it requires. Give one test with a pass criterion for each procedure.